The coordinate Riesz transform for a separated Schrödinger operator

A full answer to a concrete unknown case in arXiv:1503.02508

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Status. Candidate full solution, likely valid, pending human review. The concrete partial Riesz transform singled out as unknown in the source is bounded on $L^p(\mathbb{R}^2)$ for every $1 < p < \infty$. This answers that example, not the source's general Question A.

1 Source question

Bernicot and Frey consider a nonnegative potential V on $\mathbb{R}$ and

$$L = -\Delta + V(x_2) = \left(-\partial_{x_1}^2\right) \otimes I + I \otimes \left(-\partial_{x_2}^2 + V(x_2)\right)$$

on $L^2(\mathbb{R}^2)$. On source PDF page 8 they state that, for $\Gamma = \partial_{x_1}$, boundedness of $\Gamma L^{-1/2}$ is unknown, even though the semigroup gradient estimate (G_p) holds for all p [1].

- **Examples where (G_p) is known.**
  We shall give here two examples of situations where (G_p) is known for some $p > 2$. However (R_p) is still unknown, which motivates to ask Question A.
  (a) Consider V a non-negative potential on $\mathbb{R}$ and define on $\mathbb{R}^2$ (equipped with its Euclidean structure), the operator for $x = (x_1, x_2) \in \mathbb{R}^2$

  $$Lf(x) = -\Delta f(x) + V(x_2)f(x).$$

  Then L generates a semigroup, which has a heat kernel satisfying Gaussian bounds. For $\Gamma = \partial_{x_1}$, since V is very nice along the first coordinate, it is natural to ask for the boundedness of $\Gamma L^{-1/2}$. Such a boundedness is unknown and does not seem to be trivial, since $L^{-1/2}$ makes interact the action of L on the two coordinates. However, it is surprising to observe that (G_p) easily holds for every $p \in [1, \infty]$. Indeed, since $L_1 := -\partial_{x_1}^2$ and $L_2 := -\partial_{x_2}^2 + V(x_2)$ commute, we then have $e^{-tL} = e^{-tL_1} \otimes e^{-tL_2}$ such that

  $$\|\sqrt{t}\Gamma e^{-tL}\|_{p\to p} \leq \|\sqrt{t}\partial_{x_1} e^{-tL_1}\|_{p\to p}\|e^{-tL_2}\|_{p\to p}$$

The same paper recalls that its Gaussian/off-diagonal framework already gives boundedness of the Riesz transform on L^q for every $1 < q < 2$. In this example the heat kernel has Gaussian bounds, so the lower endpoint is one.

2 Proof intuition

The extra structure of this example makes the lower-p theorem self-dual. The derivative ∂_{x_1} is skew-adjoint and commutes with L, because the potential is independent of x_1. Hence

$$T = \partial_{x_1} L^{-1/2} \qquad \text{satisfies} \qquad T^* = -T$$

on L^2. If $p > 2$, its conjugate exponent p' lies below two, where the source's general theory already bounds T. The skew-adjoint identity then transfers that estimate to L^p by ordinary duality.

3 A duality lemma

Lemma 1. *Let T be bounded on $L^2(X)$ and suppose a dense common core satisfies*

$$\langle Tf, g\rangle = -\langle f, Tg\rangle.$$

If T extends boundedly to $L^q(X)$ for some $1 < q < 2$, then it extends boundedly to $L^{q'}(X)$ and

$$\|T\|_{L^{q'} \to L^{q'}} \leq \|T\|_{L^q \to L^q}.$$

Proof. For core functions f and g,

$$|\langle Tf, g\rangle| = |\langle f, Tg\rangle| \leq \|f\|_{q'} \, \|Tg\|_q \leq \|T\|_{q\to q} \, \|f\|_{q'} \, \|g\|_q.$$

Taking the supremum over g of L^q norm one gives the claimed bound on the core; density gives the extension. □

4 Resolution of the source example

Theorem 2. *Let $V \geq 0$ be as in the source example, so that $L = -\Delta + V(x_2)$ is the nonnegative self-adjoint Schrödinger operator on $L^2(\mathbb{R}^2)$ and its heat semigroup has the stated Gaussian bounds. Then*

$$\partial_{x_1} L^{-1/2} : L^p(\mathbb{R}^2) \longrightarrow L^p(\mathbb{R}^2)$$

is bounded for every $1 < p < \infty$.

Proof. First, the quadratic-form inequality

$$\|\partial_{x_1} u\|_2^2 \leq \|\nabla u\|_2^2 + \int_{\mathbb{R}^2} V(x_2)|u(x)|^2 \, dx = \|L^{1/2}u\|_2^2$$

shows that $T = \partial_{x_1} L^{-1/2}$ is bounded on L^2.

Translations in the first coordinate commute with L: both $-\partial_{x_1}^2$ and $-\partial_{x_2}^2 + V(x_2)$ do. Their infinitesimal generator ∂_{x_1} therefore strongly commutes with the spectral calculus of L. On $C_c^\infty(\mathbb{R}^2)$, and hence as an L^2 operator,

$$T^* = (\partial_{x_1} L^{-1/2})^* = -L^{-1/2}\partial_{x_1} = -\partial_{x_1} L^{-1/2} = -T.$$

The source's recalled low-exponent Riesz-transform theorem, applied to its Gaussian heat kernel, gives boundedness of T on L^q for every $1 < q < 2$. Lemma 1 gives boundedness on $L^{q'}$ for every $2 < q' < \infty$. Together with the L^2 estimate, this covers all $1 < p < \infty$. □

Corollary 3 (Separated-coordinate extension)**.** *Let $x = (x_j, x') \in \mathbb{R} \times \mathbb{R}^{n-1}$ and $L = -\Delta + V(x')$ with $V \geq 0$. Whenever the standard low-exponent Riesz-transform estimate holds for this Schrödinger operator, the partial transform $\partial_{x_j} L^{-1/2}$ is bounded on every L^p, $1 < p < \infty$.*

Proof. The derivative in the absent coordinate is skew-adjoint and commutes with L. Apply the same L^2 form estimate and Lemma 1. □

5 Upgrade attempts, novelty, and scope

A first route treats the transform as an operator-valued Fourier multiplier in x_1; this requires uniform functional-calculus estimates for $-\partial_{x_2}^2 + V$ and obscures the simple mechanism. The decisive upgrade is to combine the already-known $p < 2$ estimate with commutation and skew-adjointness. The same argument yields the separated-coordinate corollary above.

A bounded primary-source search covered the exact operator and potential, "partial Riesz transform", "Schrödinger", the source title and authors, and later arXiv references. It found the source and work on different localized or degenerate partial transforms, but no primary paper stating this answer for the displayed separated model. Novelty confidence is moderate because the duality argument is elementary; correctness confidence is high.

The proof is special to a linear skew-adjoint component Γ commuting with L. It does not answer whether (G_p) implies (R_p) for a general gradient operator, nor does it settle the source's second example with variable complex coefficients. Human review should verify the identification of the source's $p < 2$ theorem with the scalar partial transform and the strong commutation step on the chosen operator domains.

References

[1] F. Bernicot and D. Frey, *Riesz transforms through reverse Hölder and Poincaré inequalities*, arXiv:1503.02508.